

Increasing Logarithmic Weights for Rolling Validation: Simulation Study

Aileen Tao

1 Purpose

The companion theory report, *Increasing Logarithmic Weights for Rolling Validation: Theory* (08A), derives asymptotic expressions for the expectation, variance, and Chebyshev ratio of several increasing logarithmic weighting schemes. This report examines whether those patterns are visible at finite sample sizes and whether the empirical wrong-selection probability decreases accordingly.

For $Y_i = \theta_\star + \xi_i$ with independent standard Gaussian ξ_i , 08A compares the pre-validation sample mean with the oracle $\theta_\star$. Write Ξ_n for the weighted difference between their squared validation losses. A positive value favors the oracle, whereas $\Xi_n < 0$ is an incorrect selection. The weights appear throughout in the same order:

1. 1;
2. $\log(i + 1)$;
3. $[\log(i + 1)]^2$;
4. $1 + \log(i + 1) + [\log(i + 1)]^2$;
5. $\log \log(i + e)$.

The unweighted choice is a reference case. The remaining four weights increase, at different rates.

2 Simulation design

The validated cached experiment contains 5,000 independent replications, seed 2026, processed in batches of 250. The sample sizes are 100, 200, 500, 1000, 2000, 5000, 10000, 20000, 50000, 100000. Each Gaussian path is reused across weights and sample sizes, so comparisons are dependent. Exact moment trajectories extend from $n = 50$ through 10^7 on an approximately log-spaced grid that includes the simulation sizes.

The original implementation checks the quadratic-form representation, exact moments, direct score calculations, and the variance recurrence. The largest absolute discrepancy across these checks is $5.7\text{e-}14$. This report reads those cached results for the five weights above. The design fixes the comparison before examining the moment, ratio, and selection plots.

Table 1: Simulation design for the five weights.

Item	Value
Replications	5,000
Seed	2026
Batch size	250
Simulation sample sizes	100 to 100,000 (10 specified sizes)
Exact-trajectory range	50 to 10,000,000
Path reuse	Across weights and sample sizes

3 Validation of theoretical moments

Before interpreting selection errors, we check whether simulated moments match the exact moments used in 08A. Separate vertical scales accommodate the different magnitudes of the five weights.

3.1 Expectation

08A predicts mean growth of order $\log n$ for the baseline, $(\log n)^2$ and $(\log n)^3$ for the two log powers, $(\log n)^3$ for the polynomial, and $(\log n) \log \log n$ for the log-log weight. Figure 1 compares exact expectations with Monte Carlo means.

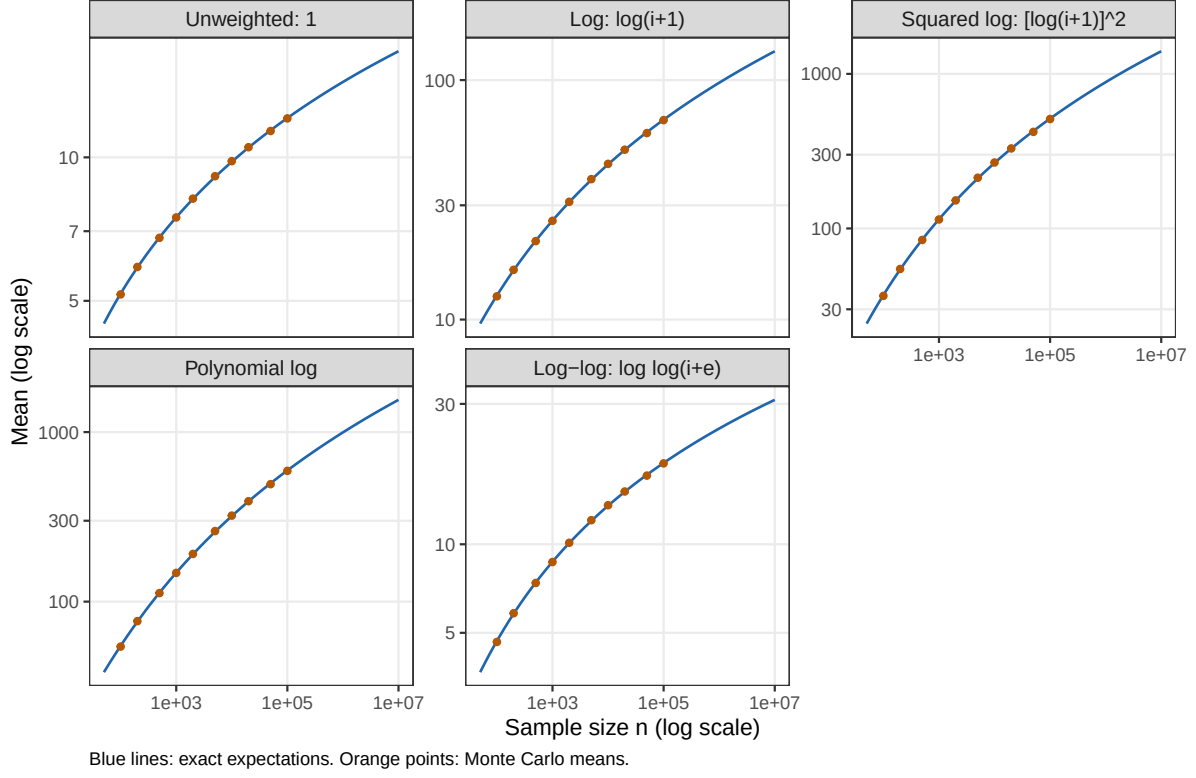


Figure 1: Exact and empirical expectations for the five weights.

The empirical points follow the exact curves across the simulated sizes. In particular, the polynomial and squared-log means approach the same leading scale, as the highest-degree-term result predicts.

3.2 Variance

08A predicts variance tending to π^2 for the baseline, growing as $2(\log n)^2$ and $2(\log n)^4$ for the log powers and polynomial, and as $2(\log \log n)^2$ for the log-log weight. Figure 2 compares those exact variance trajectories with empirical variances.

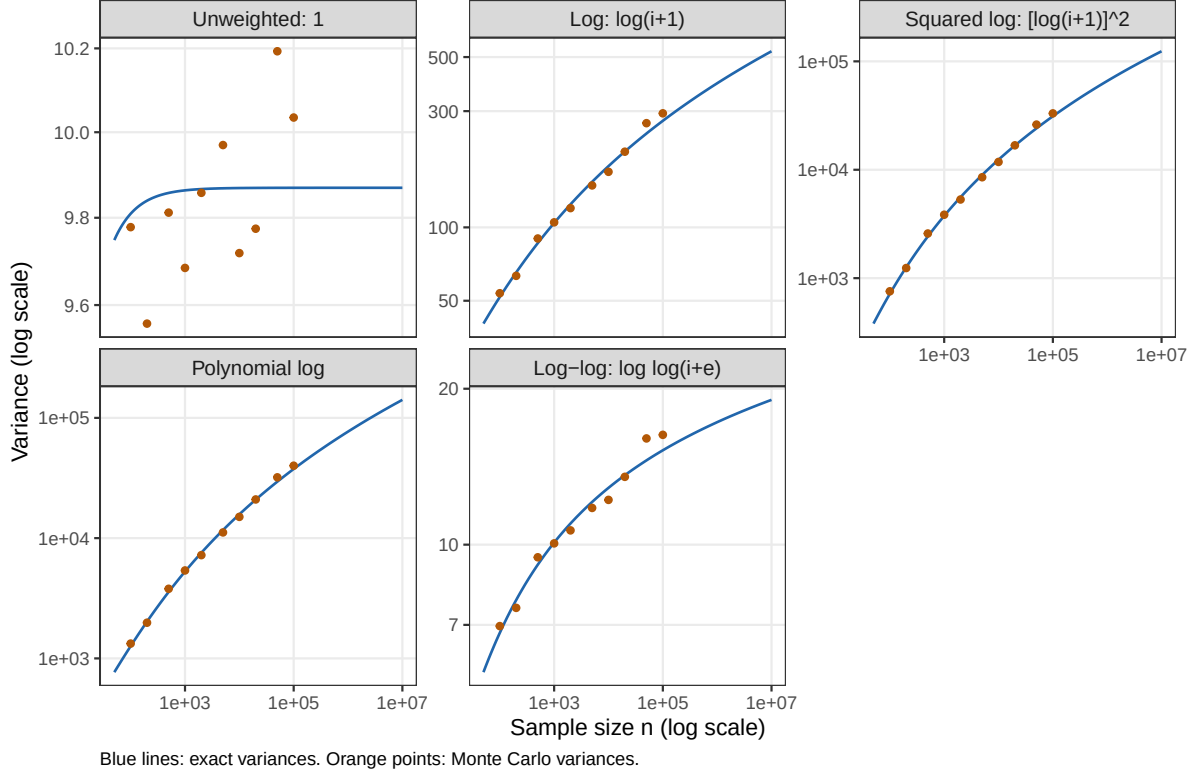


Figure 2: Exact and empirical variances for the five weights.

The empirical variances fluctuate more than the means but track the exact curves. The squared-log and polynomial cases show much larger variability over this range, even though their leading asymptotic forms agree. These moment checks support the calculations used in 08A and lead to the ratio comparison below.

4 Chebyshev ratio

08A predicts a ratio asymptotic to a weight-specific constant divided by $(\log n)^2$ for each choice. Figure 3 examines whether that downward pattern is visible at finite n . The blue curves use exact moments; the orange points use the Monte Carlo variance and exact mean.

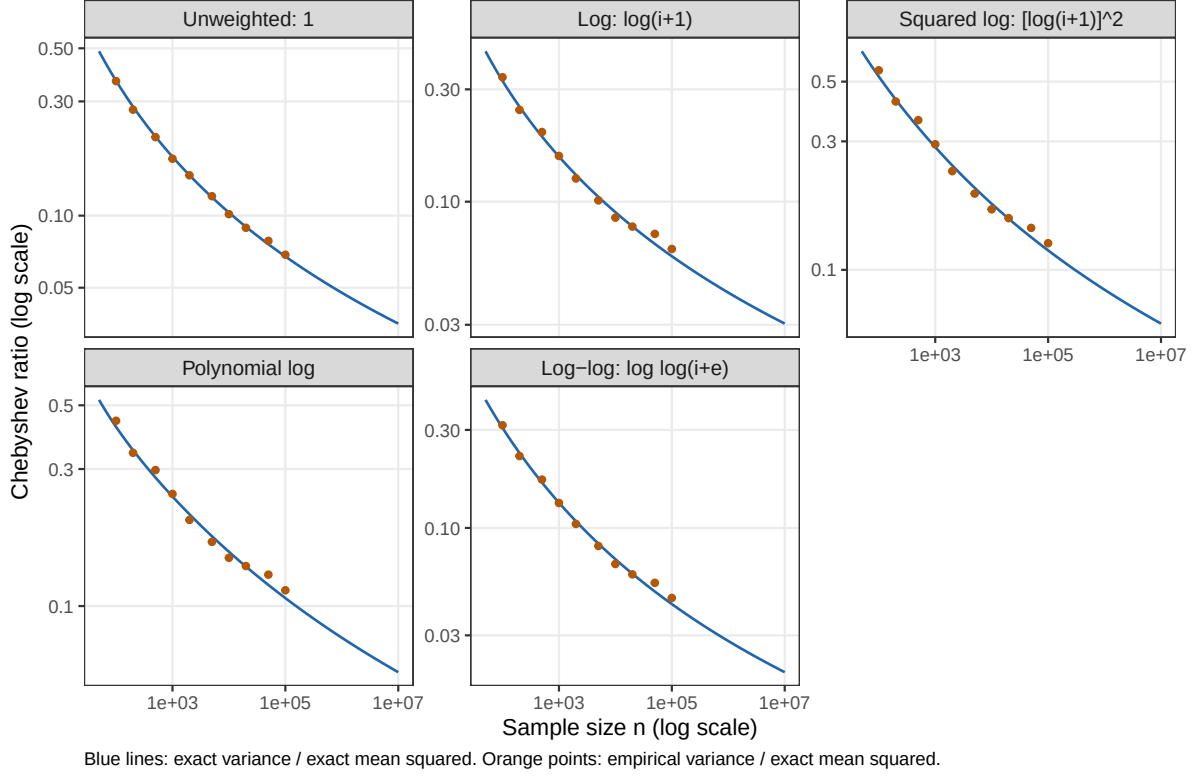


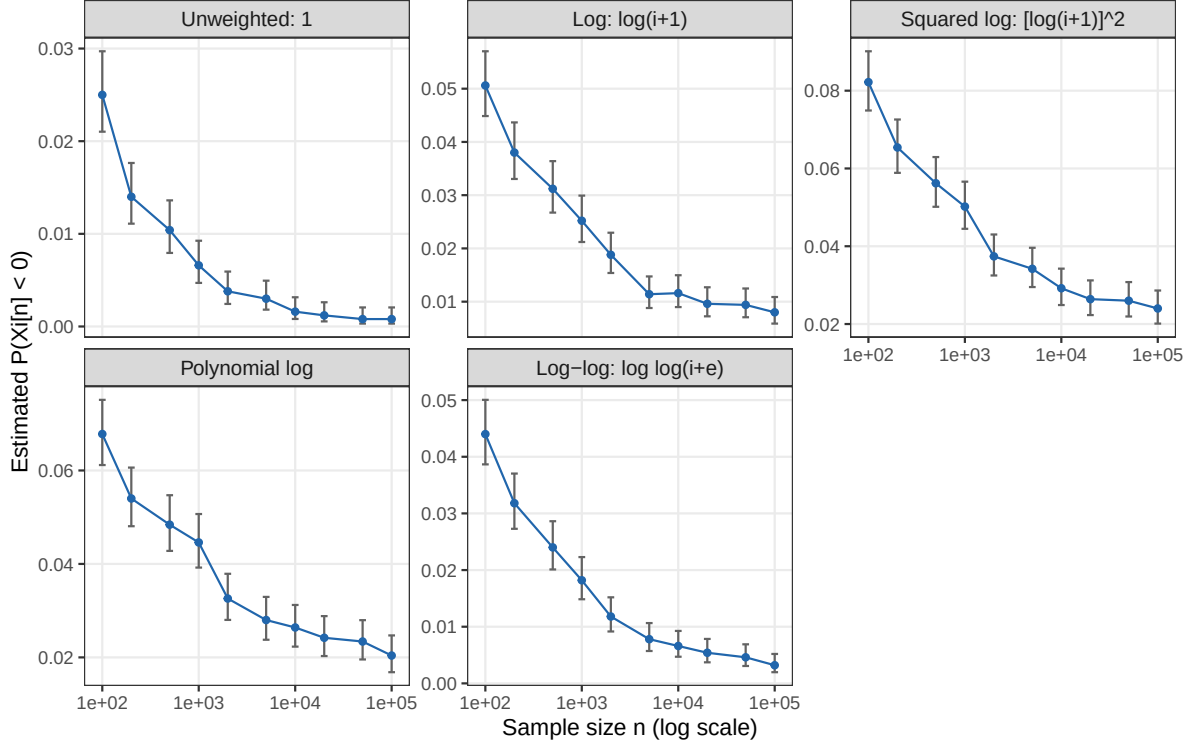
Figure 3: Exact and empirical Chebyshev ratios.

Every exact ratio decreases over the displayed range, and the empirical points follow the same broad pattern. This is consistent with the predicted $(\log n)^{-2}$ asymptotic rate, though these finite curves alone do not establish that rate. The squared-log and polynomial-log curves remain different at finite n because the polynomial contains lower-degree terms. The log-log ratio also decreases despite its very slow weight growth.

The ratio is a sufficient bound for consistency; the next section examines the actual selection errors separately.

5 Wrong-selection probability

08A shows that the Chebyshev ratio tends to zero, which is sufficient for $\mathbb{P}(\Xi_n < 0) \rightarrow 0$. Figure 4 instead estimates that probability directly from the simulated fraction of incorrect selections. The ratio is a theoretical upper-bound quantity; the frequency measures actual finite-sample errors. Bars are pointwise 95% Wilson intervals.



Pointwise 95% Wilson intervals from 5,000 replications. Shared paths make estimates across weights and n dependent.

Figure 4: Wrong-selection frequencies and pointwise Monte Carlo uncertainty.

Wrong-selection frequencies decline for every weight, as the consistency results suggest. The rate of finite-sample improvement differs: the decline is slowest for the squared-log and polynomial-log choices over this range, consistent with their larger variability. The log and log-log frequencies also fall, while the unweighted case is a low-error reference. Thus faster weight growth does not automatically improve finite-sample selection.

Table 2: Exact moments, Monte Carlo moments, ratios, and wrong-selection frequencies at $n = 100,000$.

Weight	Exact mean	MC mean	Exact variance	MC variance	Exact ratio	Wrong-selection frequency	Wilson lower	Wilson upper
Unweighted: 1	12.0901	12.0707	9.8695	10.0347	0.0675	0.0008	0.0003	0.0021
Log: $\log(i+1)$	68.3192	68.0434	273.7620	293.8490	0.0587	0.0080	0.0059	0.0109
Squared log: $[\log(i+1)]^2$	513.4996	510.3664	31143.6081	33119.3697	0.1181	0.0240	0.0201	0.0286
Polynomial log	593.9089	590.4804	37596.7760	40033.3312	0.1066	0.0204	0.0168	0.0247
Log-log: $\log \log(i+e)$	18.8973	18.8386	15.2116	16.2936	0.0426	0.0032	0.0020	0.0052

6 Weight-specific interpretation

The moment, ratio, and error plots can now be read together for each increasing weight.

Unweighted baseline. The variance approaches a finite limit, and the wrong-selection frequency declines across the simulated sizes.

Log weight. Its exact mean and variance grow on the scales predicted in 08A, while both the Chebyshev ratio and wrong-selection frequency decline.

Squared-log weight. The empirical moments follow the exact trajectories. The ratio and error frequency decrease more slowly at the available sample sizes than for the log weight, but the direction agrees with the theory.

Polynomial-log weight. Its trajectories resemble those of squared log, while retaining visible finite-sample differences from the lower-degree terms. This is the expected approach to highest-degree-term dominance.

Log-log weight. Both the ratio and wrong-selection frequency decline even though the weight grows extremely slowly, matching the qualitative conclusion of 08A.

7 Summary

Across the five weights, the empirical moments track the exact calculations, the ratios decline in the direction predicted by 08A, and wrong-selection frequencies decrease over the simulated sizes. The squared-log and polynomial-log cases improve more slowly at finite n , while the log-log case also shows the predicted downward behavior.

These numerical results do not replace the asymptotic proofs in 08A. They show how the theoretical mechanisms appear at finite sample sizes and preserve the distinction between a sufficient Chebyshev argument and the directly observed wrong-selection frequency.